A GESTURE CONTROLLED 3D PLATFORMER GAME

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Game Concept

This project is a **3D platformer** developed in **Unity 3D**, where the primary objective is to navigate through levels and reach the endpoint. The game introduces a unique **time-manipulation mechanic**, allowing players to rewind and fast-forward time to solve environmental puzzles and overcome obstacles.

Core Mechanics

Player Movement & Time Control

- Movement (Right Hand Tracking):

  - The player's movement is controlled via **hand gestures** detected using computer vision.

  - Thumb direction determines movement:

    - Up → Forward

    - Down → Backward

    - Left → Move Left

    - Right → Move Right

First run the python code and then start the game in unity editor.

- **Time Manipulation (Left Hand Tracking):**

  - The player can control the flow of time using gestures:

    - Thumb Right → Fast-Forward Time

    - Thumb Left → Rewind Time

Technical Implementation

Computer Vision & Input System

- **Model Used**: MediaPipe’s pre-trained hand landmark detection model (approved by competition organizers).

- **Gesture Recognition**:

  - A coordinate-based logic system computes the thumb’s direction in real-time.

  - Eliminates the need for frame-by-frame machine learning predictions, reducing latency.

- **Data Pipeline:**

  - The Python-based hand-tracking system processes live camera input and detects gestures.

  - Detected inputs are transmitted to Unity via UDP (User Datagram Protocol) for seamless integration.

Machine Learning Exploration

- A **custom CNN model** was initially trained using the [Hand Navigation Landmarks Dataset] (https://www.kaggle.com/datasets/harisudhan411/hand-navigation-landmarks) to classify thumb directions.

- Challenges:

  - Per-frame prediction introduced **significant computational overhead**.

  - Increased input lag, negatively impacting gameplay responsiveness.

- Solution:

  - Switched to MediaPipe's lightweight landmark detection with coordinate-based logic for real-time performance.

Conclusion

This game combines innovative gesture-based controls with dynamic time manipulation, offering a unique puzzle-platformer experience. By leveraging MediaPipe's efficient hand-tracking model and a UDP-based input system, the game ensures low-latency, responsive controls while maintaining an engaging gameplay loop.